

When studying human memory, psychologists use a paradigm called free recall. In a free recall task, a subject is presented a list of to-be-recalled items, one after another. For example, an experimenter might read off a list of 15 words, presenting a new word to the test taker every 5 seconds. When the experimenter has read the entire list, the subject is asked to recall as many of the listed items as possible (e.g., by writing them down). This is known as a *free recall task* because the subject is free to recall the items in any order that he or she desires.

The results of a free recall task are plotted on something called a *serial position curve*, which is normally U-shaped. The serial position curve is graphed on a basic, coordinate plane, in which the x axis plots the serial position of to-be-remembered items in the list (e.g., the first item, the second item, the third item, and so on). The y axis, meanwhile, indicates the probability of recalling the item, which is based on the average frequency of recall across a number of subjects in a given trial. The serial position curve, once constructed, tends to exhibit both a recency and a primacy effect.

The primacy effect describes the left portion of the U shape, that is, the elevated portion at the beginning of the curve, which precedes the concavity at the middle. The primacy effect dictates that, in a free recall experiment like the one just described, articles presented at the beginning of a list of to-be-remembered items are remembered better than ones presented in the middle of the list. The primacy effect appears to result from subjects recalling items directly from semantic memory—a type of memory that might be thought of as the “hard drive” of a computer brain. The first items inscribed in a given session onto this hard drive are more precisely retained and are easier to access than items inscribed later on.

The recency effect describes the right portion of the serial position curve. When the recency effect appears in a free recall experiment, it means that subjects recall items appearing toward the end of the to-be-remembered list better than they remember items appearing in the middle. The effect is named in such a way because the observations comprising the right-hand tail of the serial position curve correspond to the items the subjects heard most recently prior to the recall challenge. Recency bias is the result of subjects recalling items directly from short-term memory. In continuing the computer analogy, if semantic memory represents a portion of your brain’s long-term memory, or “hard drive,” then short-term memory is like random access memory (RAM), which contains data that your computer can access dynamically during a session, but which it may also lose after rebooting. Short-term memory stores only limited quantities of information over limited periods of time. Therefore, while the primacy effect results from the extra-long-term memory rehearsal accorded to primary items on a list, the recency effect occurs because the items the subject heard most recently are more likely to